II-5

$$OP = |OP|\ \mathrm{dir}(OP), \qquad \mathrm{dir}(OP) = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$$

$$|OP| = |OQ| + |QP| = a + a\theta$$

$$OP = \mathbf{r} = a(\cos\theta + \theta\sin\theta)\mathbf{i} + a(\sin\theta - \theta\cos\theta)\mathbf{j}$$

II-6

$$r_1(t) = (10 - t)\mathbf{i} \qquad \text{(hunter)}$$

$$r_2(t) = t\mathbf{i} + 2t\mathbf{j} \qquad \text{(rabbit)}$$

$$v = \mathbf{i} + 2\mathbf{j}$$

$$|v| = \sqrt{5}$$

$$\text{Arrow} = tA = \frac{r_2 - r_1}{|r_2 - r_1|}$$

$$= \frac{(t-5)\mathbf{i} + t\mathbf{j}}{\sqrt{2t^2 - 10t + 25}}$$

Let

$$f(t) = (r_2 - r_1)^2 = 2t^2 - 10t + 25$$

$$f'(t) = 4t - 10 = 0 \quad \text{when } t = 2.5$$

II-7

$$OP = OA + AB + BP$$

$$OA = a\,\mathrm{arc}AP = a\theta$$

$$AB = a\mathbf{j}$$

$$BP = a(-\sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{j})$$

1J-1
a)

$$\mathbf{r} = e^t\mathbf{i} + e^{-t}\mathbf{j}$$

$$\mathbf{v} = e^t\mathbf{i} - e^{-t}\mathbf{j}$$

$$|v| = \sqrt{e^{2t} + e^{-2t}}$$

$$\mathbf{T} = \frac{e^t\mathbf{i} - e^{-t}\mathbf{j}}{\sqrt{e^{2t} + e^{-2t}}}$$

$$\mathbf{a} = e^t\mathbf{i} + e^{-t}\mathbf{j}$$

b)

$$\mathbf{r} = t^2\mathbf{i} + t^3\mathbf{j}$$

$$\mathbf{v} = 2t\mathbf{i} + 3t^2\mathbf{j}$$

$$|v| = t\sqrt{4 + 9t^2}$$

$$\mathbf{T} = \frac{2\mathbf{i} + 3t\mathbf{j}}{\sqrt{4 + 9t^2}}$$

$$\mathbf{a} = 2\mathbf{i} + 6t\mathbf{j}$$

c)

$$\mathbf{r} = (1 - 2t^2)\mathbf{i} + t\mathbf{j} + (2 + 2t^2)\mathbf{k}$$

$$\mathbf{v} = 2t(-2\mathbf{i} + \mathbf{j} + 2\mathbf{k})$$

$$|v| = 6t$$

$$\mathbf{T} = \frac{1}{3}(-2\mathbf{i} + \mathbf{j} + 2\mathbf{k})$$

$$\mathbf{a} = 2(-2\mathbf{i} + \mathbf{j} + 2\mathbf{k})$$

1J-2 (a)

$$\mathbf{r} = \frac{1}{1 + t^2}\mathbf{i} + \frac{t}{1 + t^2}\mathbf{j}$$

$$\mathbf{v} = \frac{-2t}{(1+t^2)^2}\mathbf{i} + \frac{1-t^2}{(1+t^2)^2}\mathbf{j}$$

$$|v| = \frac{1}{1+t^2}$$

$$\mathbf{T} = \frac{-2t\mathbf{i} + (1-t^2)\mathbf{j}}{1+t^2}$$

1J-3

$$\frac{d}{dt}(x_1y_1 + x_2y_2) = x_1'y_1 + x_1y_1' + x_2'y_2 + x_2y_2'$$

Adding the columns we get

$$(x_1, x_2)'(y_1, y_2$$

1J-4

a) Since P moves on a sphere of radius a, we have

$$x(t)^2 + y(t)^2 + z(t)^2 = a^2$$

Differentiating with respect to t,

$$2xx' + 2yy' + 2zz' = 0$$

Hence

$$xx' + yy' + zz' = 0$$

or in vector form

$$\mathbf{r} \cdot \mathbf{r}' = 0$$

b) Since by hypothesis $|\mathbf{r}(t)| = a$, for all t we have

$$\mathbf{r} \cdot \mathbf{r} = a^2$$

Differentiating,

$$\frac{d}{dt}(\mathbf{r} \cdot \mathbf{r}) = 2\mathbf{r} \cdot \mathbf{r}' = 0$$

Hence

$$\mathbf{r} \cdot \mathbf{v} = 0$$

c) Using the result above,

$$\mathbf{r} \cdot \mathbf{r} = |\mathbf{r}|^2$$

If

$$\mathbf{r} \cdot \mathbf{v} = 0$$

then

$$\frac{d}{dt}(\mathbf{r} \cdot \mathbf{r}) = 0$$

so

$$\mathbf{r} \cdot \mathbf{r} = c$$

Therefore

$$|\mathbf{r}| = \sqrt{c}$$

which shows that the head of $\mathbf{r}$ moves on a sphere of radius $\sqrt{c}$.

1J-5
a) If

$$|\mathbf{v}| = c$$

then

$$\mathbf{v} \cdot \mathbf{v} = c^2$$

Differentiating,

$$\frac{d}{dt}(\mathbf{v} \cdot \mathbf{v}) = 2\mathbf{v} \cdot \mathbf{a}$$

Hence

$$\mathbf{v} \cdot \mathbf{a} = 0$$

so velocity and acceleration vectors are perpendicular.

b) If

$$\mathbf{v} \cdot \mathbf{a} = 0$$

then

$$\frac{d}{dt}(\mathbf{v} \cdot \mathbf{v}) = 0$$

so

$$|\mathbf{v}| = \sqrt{a}$$

which shows the speed is constant.
1J-6
a)

$$r = a\cos t\,\mathbf{i} + a\sin t\,\mathbf{j} + bt\,\mathbf{k}$$

$$v = -a\sin t\,\mathbf{i} + a\cos t\,\mathbf{j} + b\,\mathbf{k}$$

$$|v| = \sqrt{a^2 + b^2}$$

$$T = \frac{v}{\sqrt{a^2+b^2}}, \qquad a = -a(\cos t\,\mathbf{i} + \sin t\,\mathbf{j})$$

b) By direct calculation using the components we see that

$$v \cdot a = 0$$

This also follows theoretically from the exercise since the speed is constant.

1J-7

a) The criterion is $|V| = 1$. If we measure length s so that $s = 0$ when $t = 0$, and since s increases with t,

$$|V| = 1 \Rightarrow \frac{ds}{dt} = 1$$

$$s = t + C$$

When $t = 0$, $s = 0$, therefore

$$s = t$$

b)

$$r = (x_0 + at)\mathbf{i} + (y_0 + at)\mathbf{j}$$

$$V = a(\mathbf{i} + \mathbf{j})$$

$$|V| = a\sqrt{2}$$

Choose

$$a = \frac{1}{\sqrt{2}}$$

c) Choose a and b to be non–negative numbers such that

$$a^2 + b^2 = 1$$

then

$$|V| = 1$$

1J-8

a)
Let

$$r = x(t)\mathbf{i} + y(t)\mathbf{j}$$

$$u(t)r(t) = ux\,\mathbf{i} + uy\,\mathbf{j}$$

Differentiating gives

$$(ur)' = (ux)'\mathbf{i} + (uy)'\mathbf{j}$$

(ur)'

$$(ur)' = (ux\mathbf{i} + uy\mathbf{j})' = (ux)'\mathbf{i} + (uy)'\mathbf{j}$$

$$= (u'x + ux')\mathbf{i} + (u'y + uy')\mathbf{j}$$

$$= u'(x\mathbf{i} + y\mathbf{j}) + u(x'\mathbf{i} + y'\mathbf{j}) = u'r + ur'$$

b)

$$\frac{d}{dt}\left[e^t(\cos t\,\mathbf{i} + \sin t\,\mathbf{j})\right]$$

$$= e^t(\cos t\,\mathbf{i} + \sin t\,\mathbf{j}) + e^t(-\sin t\,\mathbf{i} + \cos t\,\mathbf{j})$$

$$= e^t[(\cos t - \sin t)\mathbf{i} + (\sin t + \cos t)\mathbf{j}]$$

$$|V| = e^t$$

1J-9
a)

$$r = 3\cos t\,\mathbf{i} + 5\sin t\,\mathbf{j} + 4\cos t\,\mathbf{k}$$

$$|r| = \sqrt{25\cos^2 t + 25\sin^2 t} = 5$$

b)

$$V = \frac{dr}{dt} = -3\sin t\,\mathbf{i} + 5\cos t\,\mathbf{j} - 4\sin t\,\mathbf{k}$$

$$|V| = \sqrt{25\cos^2 t + 25\sin^2 t} = 5$$

c)

$$a = \frac{dV}{dt} = -3\cos t\,\mathbf{i} - 5\sin t\,\mathbf{j} - 4\cos t\,\mathbf{k}$$

d) By inspection the coordinates x, y, z of P satisfy

$$4x - 3z = 0$$

which is a plane through the origin.

e) Since by part (a) the point P moves on the surface of a sphere of radius 5 centered at the origin, and by part (d) also lies in the plane $4x - 3z = 0$, its path is the intersection of these two surfaces, which is a great circle of radius 5 on the sphere.

1J-10

$$T = \frac{-a\sin t\,\mathbf{i} + a\cos t\,\mathbf{j} + b\,\mathbf{k}}{\sqrt{a^2+b^2}}, \qquad |V| = \sqrt{a^2+b^2}$$

By the chain rule

$$\left|\frac{dT}{dt}\right| = \left|\frac{dT}{ds}\right|\left|\frac{ds}{dt}\right|$$

Therefore

$$\left|\frac{-a\sin t\,\mathbf{i} + a\cos t\,\mathbf{j}}{\sqrt{a^2+b^2}}\right| = k\sqrt{a^2+b^2}$$

Since the numerator on the left has value $|a|$, we get

$$k = \frac{|a|}{a^2+b^2}$$

b) If $b = 0$, the helix is a circle of radius $|a|$ in the xy–plane and

$$k = \frac{1}{|a|}$$

1K-1

$$\frac{d}{dt}(x_1y_1 + x_2y_2) = x_1'y_1 + x_1y_1' + x_2'y_2 + x_2y_2'$$

Adding the columns gives

$$(x_1, x_2)' \cdot (y_1, y_2) + (x_1, x_2) \cdot (y_1, y_2)' = \frac{d}{dt}(x \cdot y)$$

1K-2

In two dimensions

$$s(t) = (x(t), y(t)), \qquad s'(t) = (x'(t), y'(t)).$$

Therefore

$$s'(t) = 0 \Rightarrow x'(t) = 0,\ y'(t) = 0$$

$$\Rightarrow x(t) = k_1,\ y(t) = k_2$$

$$\Rightarrow s(t) = (k_1, k_2)$$

where k_1, k_2 are constants.

1K-3

Since F is central, we have

$$F = cr$$

Using Newton's law

$$a = \frac{F}{m} = \frac{c}{m}r$$

$$F = cr \Rightarrow r \times a = r \times \frac{c}{m}r = 0$$

$$\Rightarrow \frac{d}{dt}(r \times v) = r \times a = 0$$

$$\Rightarrow r \times v = K \quad \text{(a constant vector)}$$

$$|r \times v| = 2\frac{dA}{dt}$$

$$\frac{dA}{dt} = \frac{1}{2}|K|$$

which shows the area is swept out at a constant rate.